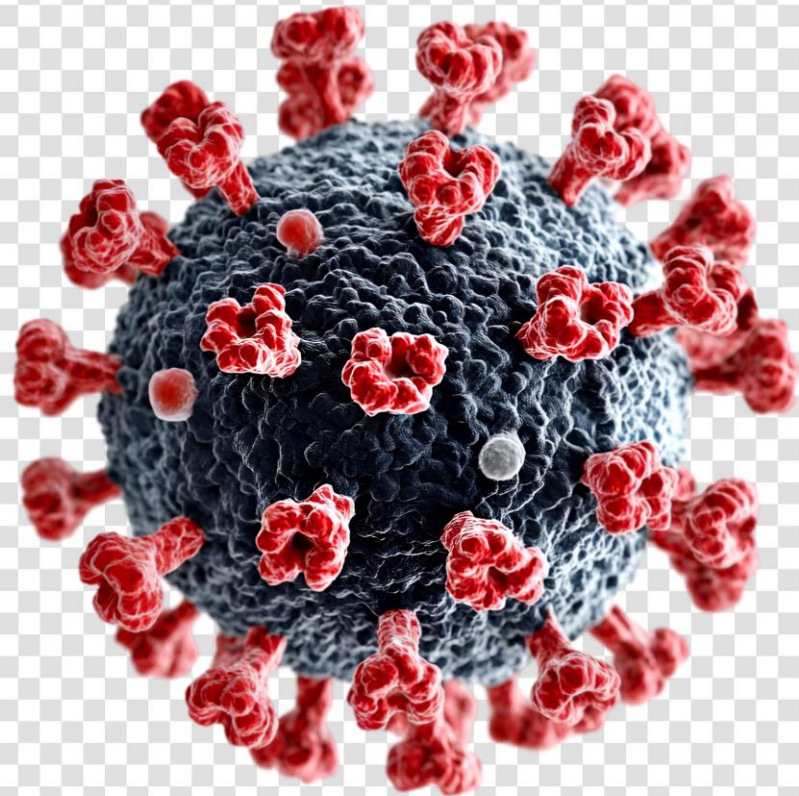
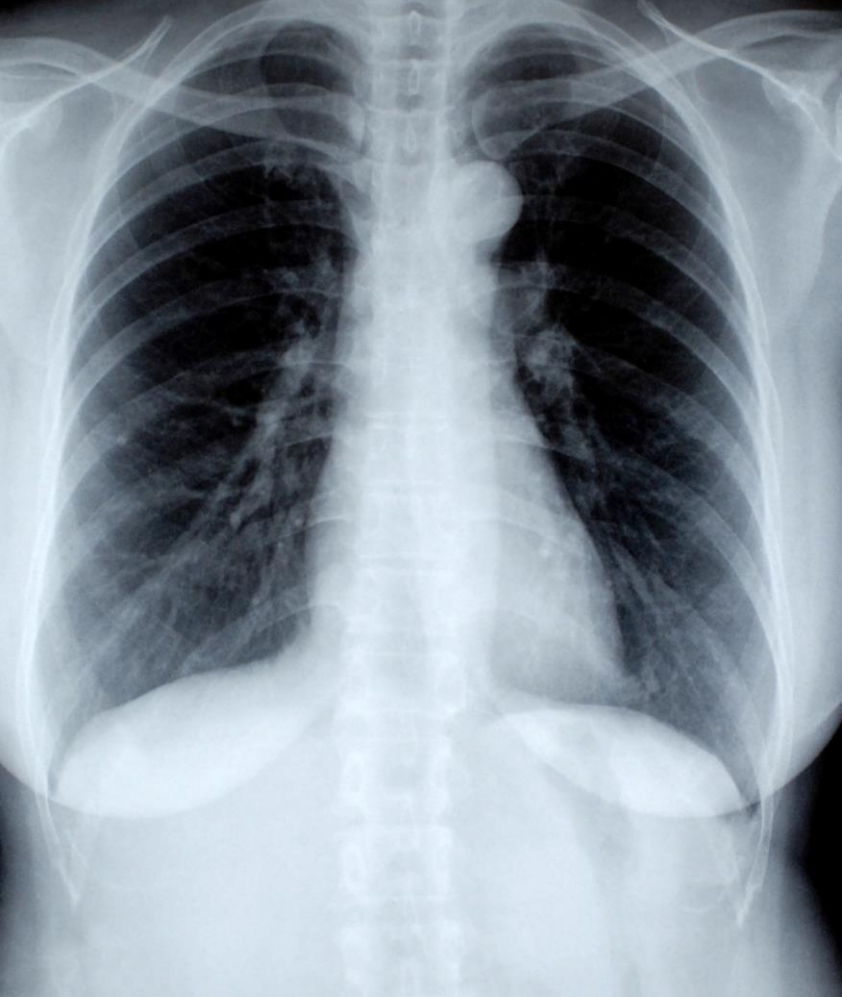




LUNG CANCER RISK PREDICTION USING MACHINE LEARNING

TOOLKITS: PYTHON, PANDAS, SKLEARN, XGBOOST, STREAMLIT

LUNG CANCER RISK PREDICTION PROJECT AND CONTENT LAYOUT



Project Objective



Data Overview



Explorative Data Analysis



Model Building



Model Performance



Web App Deployment



Recommendations



Conclusion



PROJECT OBJECTIVES

A machine learning project focused on predicting the risk of pulmonary disease using patient health data. This presentation summarizes key results, insights, and recommendations.

DATA OVERVIEW



Dataset consisted of demographic and health-related variables.



Key features: AGE, GENDER, SMOKING, POLLUTION EXPOSURE, STRESS, FAMILY HISTORY, etc.



Target variable: Presence of Pulmonary Disease (Yes/No).

EXPLORATIVE DATA ANALYSIS(EDA)



Visualized age distribution, categorical variables, and continuous features.

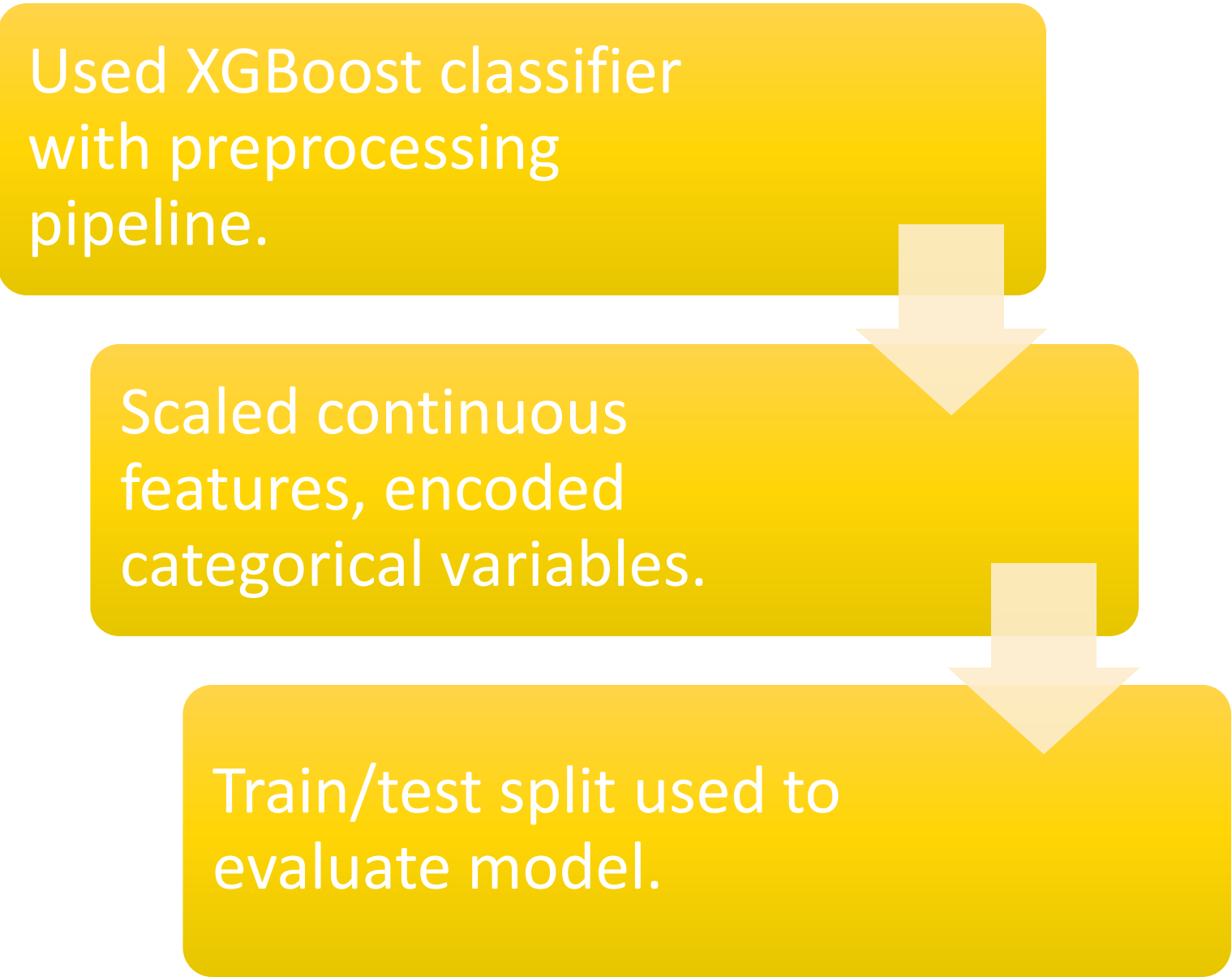


Identified trends: higher risk with increasing age, family history, smoking habits.



Handled missing values, outliers, and encoded features appropriately.

Used XGBoost classifier
with preprocessing
pipeline.



```
graph TD; A[Used XGBoost classifier with preprocessing pipeline.] --> B[Scaled continuous features, encoded categorical variables.]; B --> C[Train/test split used to evaluate model.];
```

Scaled continuous
features, encoded
categorical variables.

Train/test split used to
evaluate model.

MODEL BUILDING

MODEL PERFORMANCE



Confusion Matrix: High accuracy and precision in classification.



AUC-ROC Score: 0.92 indicating excellent discriminative power.



Final model saved as a pipeline and integrated into a web app using Streamlit.

Lung Cancer Risk Prediction App

Gender

Male

Age

29

18

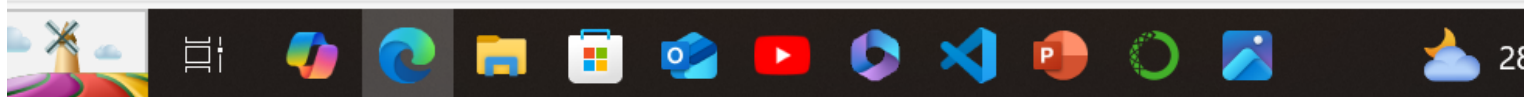
Smoking

Yes

Finger Discoloration

Yes

Mental Stress



WEB APP DEPLOYMENT

- Developed and deployed Streamlit app.
- Users input health attributes to predict lung cancer risk.
- App shows results and probability score for risk prediction.

RECOMMENDATIONS

01

Focus awareness campaigns on individuals with risk traits (e.g., smokers, elderly).

02

Integrate risk prediction into early screening programs.

03

Expand model with more diverse data for better generalization.

04

Maintain ethical use and privacy of patient data.



DATA

THANK YOU

Questions? Feedback? REACH OUT!

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